

QueryGraph with Pinax

Enterprise contracts for governed agent reads

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The stack's responsibilities

Component	Responsibility
Grust	Property graphs and graph queries
TypeSec	Identity and policy evaluation
Marciana	Semantic catalog and agent workflows
LakeCat + Sail	Catalog authority and physical execution
Pinax	Validated, versioned table contracts
QueryGraph	Trusted composition and MCP

The original semantic demo

A question connects semantic discovery to a signed agent workflow and lineage

```
querygraph dataverse-e2e --live-sail \  
  --sail-dir "$RUN/sail" \  
  --openlineage-file "$RUN/events.jsonl" \  
  --did-ledger-file "$RUN/attestations.jsonl"
```

Live Sail: two datasets, 34 graph nodes, 33 edges and appended lineage.

What Pinax adds

A reviewed contract accompanies the table through discovery, planning and result release

- Stable field IDs, types and business semantics
- Owner, steward and exact policy binding
- Compatible revisions checked against a predecessor
- Catalog schema and registry pins checked at use

A bounded request

```
{  
  "table": "rows",  
  "columns": ["id"],  
  "purpose": "analytics",  
  "limit": 10  
}
```

The host supplies identity, registry, policy and catalog credentials.

The request cannot choose a storage path or policy engine.

Checks before returning rows

Boundary	Required check
Pinax	Purpose, table and projected columns
Catalog	Exact contract and current state
LakeCat	Separate authority to execute
Sail	Exact snapshot and mandatory predicate
Release	Fresh authority and evidence integrity

The real-row experiment

Stored record	Tenant	Result
ID 1	other	Excluded by owner predicate
ID 2	acme	Returned as {"id":2}

The private email stays out of the projection

Real Iceberg metadata and Parquet files. Direct Rust MCP and the Python CLI handoff both passed on EC2.

Changes after the read

Sail reads actual rows. Authority changes. The owner discards the buffer.

Mutation	Observed outcome on grust
Purpose revoked	No protected rows released
Registry drift	No protected rows released
Snapshot drift	No protected rows released

Revision two

A reviewed optional column becomes a controlled consumer cutover

1. Close consumers for the old registry
2. Apply the reviewed target and reconcile catalog pins
3. Start consumers pinned to the target digest
4. Read real rows under revision two

Activation is an observation. The operator owns the maintenance window.

Stateless MCP · 2026-07-28

- Per-request context; no modern initialization handshake
- Central ontology discovery over stdio and HTTP
- Explicit model handles keep conversations separate
- Up to 16 requests; disconnect cancels its HTTP work

A completed response cannot be retracted. Remote HTTP work may continue after local cancellation.

Build and run on EC2

```
bash demo/pinax/deploy.sh "$DEMO_ROOT"
```

```
ssh -N -L 18081:127.0.0.1:18081 grust  
# Open http://localhost:18081
```

Our Sail source checkout and stack compile using Rust 2024.

The console and services stay available for the presentation.

Source overrides remain explicit. This correctness run makes no performance claim.

Evidence and published release

Verified	Scope
EC2 build	QueryGraph and prepared owner/engine
Live Sail	Graph load, node readback and lineage append
Governed reads	Actual rows through both MCP entry paths
Rejection	Policy, projection, drift and backend failure
Cutover	Real rows after revision-two activation

Persistent demo services are running. Pinax 0.2.0 (`pinax-registry`) is published on crates.io.

A central ontology agents actually consult

“Account number” → Customer identifier → `rows.id`

- Pinax inventories every registered table and field
- Stewards review concepts and field mappings separately
- Signed discovery filters metadata through current policy
- The agent builds its read from the resolved binding

Versioned central publication · Croissant 1.1 result data · CDIF and SKOS

Pinax

The contract, the implementation
and the operational workflow

firstpair.org

github.com/querygraph/pinax

Book, source examples and the announcing-pinax story

